

Institute of Actuaries of India

Subject SP5 –Investment and Finance

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Principles underlying the legislation / regulation of institutional investment practices:

Effective decision-making: Decisions to be taken by persons or organizations with the skills, information and resources necessary to take them effectively. [0.5]

Trustees will need to appoint investment managers with the required expertise to manage assets under their custody and investment managers must have the necessary skills and experience to handle the particular investments. [0.5]

Clear objectives: Trustees should set out an overall investment objective for the fund that: [0.5]

- represents their best judgment of what is necessary to meet the fund's liabilities
- takes account of their attitude to risk, specifically their willingness to accept under-performance due to market conditions. [0.5]

The trustees of a mature pension scheme where a high proportion of the membership are retired might adopt a low-risk investment strategy and invest mostly in government bonds. [0.5]

Objectives for the overall fund should not be expressed in terms that have no relationship to the fund's liabilities, such as performance relative to other funds or to a market index. [0.5]

Focus on asset allocation: Strategic asset allocation decisions should receive a level of attention that fully reflects the contribution they can make towards achieving the fund's investment objective. [0.5]

Expert advice: The fund should be prepared to pay sufficient fees for each service (like investment advisory, investment management etc.) to attract a broad range of potential providers and review regularly the choice of service providers. [0.5]

Explicit mandates: Trustees should make an explicit written mandate covering agreement between trustees and managers on: [0.5]

- all the mandates are coherent with the fund's aggregate objective and risk tolerances
- the managers' approach in attempting to achieve the objective
- clear time scales of measurement and evaluation [0.5]

The mandate should not exclude the use of any set of financial instruments without clear justification in the light of the specific circumstances of the fund. [0.5]

Activism: Managers should incorporate an explicit strategy on activism (intervening in company activity, in order to ensure good corporate governance), elucidating the: [0.5]

- circumstances in which they will intervene in a company,
- the approach they will use in doing so and
- how they measure the effectiveness of this strategy. [0.5]

Appropriate benchmarks: Trustees should:

- determine benchmark through an asset liability matching exercise. While setting limits on divergence from an index, ensure that they reflect the approximations involved in index construction and selection [0.5]

- consider the efficiency, liquidity and level of transaction costs in the market concerned to set both targets and risk controls. [0.5]

Performance measurement: Trustees should arrange for measurement of performance of the fund and should make formal assessment of their own procedures and decisions as trustees. [0.5]

They should also arrange for a formal assessment of performance and decision making delegated to advisers and managers. The measurement should take account of any investment constraints imposed by the trustees on the funds. [0.5]

Transparency: Set out a “Statement of Investment Principles” covering all the principles. [0.5]

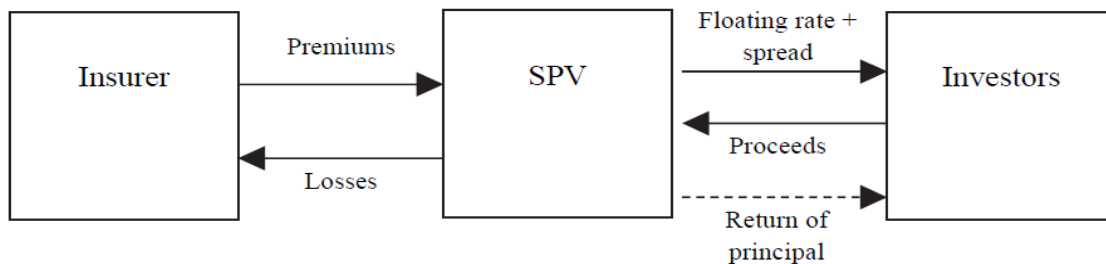
Effective operations: The selection of managers should also consider operational aspects, such as the separation of front office and back office functions. [0.5]

Regular reporting: Trustees should publish their Statement of Investment Principles and the results of their monitoring of advisers and managers, and send them annually to members of the fund. The statement should explain why the fund has decided to depart from any of these principles. [1]

[8]

Solution 2:

i) Process for creating Cat-Bonds



[2]

- The ceding insurance company establishes a special purpose vehicle in a tax efficient jurisdiction.
- The SPV establishes a reinsurance agreement with the sponsoring insurance company.
- The SPV issues a note to investors; this note has default provisions that mirror the terms of the reinsurance agreement.
- The proceeds from the note sale are invested in money market instruments within a segregated collateral account.
- If no trigger events occur during the risk period, the SPV returns the principal to investors with the final coupon payment. If trigger events occur, the assets of the SPV are first used to meet the insurer's losses, before any return of principal (if any).

[1/2 for each bullet point]

[4]

ii) Key questions regarding Cat-Bond

- What is the formula for the magnitude of the cash flows likely to arise under the Cat-Bonds?

- Whether insurer claim amounts are industry indexed or indemnity triggered
- What is the likelihood of receipt or payment of the cash flows? – Insurance event risks covered underlying risk-transfer agreement
- What is the term to maturity, yield and the credit rating?
- Any expenses involved like fee structure / brokerage / commissions
- Whether there are any callable or extendable options that exist with the issuer
- What is the extent of recourse to the issuer's assets? What is the investors ranking in case of issuers insolvency
- What is the marketability of Cat-Bonds?- liquidity in secondary markets?
- Complexity of the process of valuation - applicable regulatory valuation rules

[1/2 mark for each bullet point]

[3]

iii) ***Advantages and risks of Cat-Bonds***

Advantages of Cat-Bonds to the insurer

- Provide re-insurance protection from a new pool of capital.
- Provide diversifying sources of re-insurance capacity, thus reducing reliance on just one market
- Solves shortage/pricing issue of available traditional re-insurance capacity
- Investor capital sits in a segregated collateral account, meaning that if an event occurs, dedicated funds are available to make a payment. This virtually eliminates the credit risk inherent in traditional re-insurance
- an indemnity trigger in risk-transfer agreement will offer the lowest basis risk
- Provide multiyear pricing stability as they have typically 3-5year term whereas traditional re-insurance has annually reviewable pricing

Advantages of Cat-Bonds to the investor

- Provide stability as uncorrelated returns to that of equities and bond markets.
- Diversification as insurance events are not correlated with financial markets
- High risk adjusted returns
- Low volatility in valuations
- Provide Collateralized structures

Risks of Cat-Bonds to the insurer

- An index linked trigger in risk-transfer agreement lead to basis risk.
- Counterparty risk in case of a very large claim

Risks of Cat-Bonds to the investor

- Potentially significant insurance event risk leading to loss of all or major portion of the investment
- In callable bonds, issuer may redeem before maturity leading to loss in interest income
- Maturity may be extended by issuer without the prior consent of the investor
- Limited recourse to issuers assets and no recourse to insurer assets
- On insolvency of issuer, may lose some or all of the investment as investor ranks below the insurer
- Low marketability / liquidity leading to volatile prices in secondary markets
- Credit rating may not reflect the potential impact of all related risks

[1/2 for each bullet point]

[7]

[14 Marks]

Solution 3:**i) Approach to credit rating**

Credit rating involves a thorough investigation of the financial condition of a company in order to assess the likelihood that the company will fail to meet the interest payments and / or the repayment of principal under bonds issued. [0.5]

Rating agencies focus on:

- **Fundamental risks of the company:**

- For FMCG company
 - Demand is stable for essential items like foods, beverages, toothpastes, soaps and detergents
 - Demand for discretionary segments such as perfumes and cosmetics may be slightly cyclical
 - Product mix: popular segment targeted at low/middle income groups whereas premium segment targeted at high-income customers
- For Cement company
 - Demand is cyclical and uneven capacity additions may outpace demand during some periods. Unutilized capacities will weaken the financials
 - Location of Plant: is important as high transportation costs for both the produce and the raw materials.
 - Predominantly it is domestic demand driven.
 - Access to key inputs at competitive prices
 - Government policy on:
 - rural / urban housing and infrastructure development
 - Availability of raw materials like limestone and coal
- For both FMCG and Cement company
 - market risk (systematic) –the risks relating to the market as a whole that reflect the experience of the economy (systematic risk)
 - spread of risk – across different markets
 - event risk – exposure to specific events, such as unrelated diversification, mergers and acquisitions, business restructuring, asset sales and spin-offs, capital restructuring and litigations

- **Competitive position (relative to peers)**

- For FMCG company
 - pricing power and market share dealt with level of product differentiation and product launches by innovations or imitations for example, Colgate vedshakti in response to Patanjali products
- For Cement company
 - Consolidations via production sharing or pricing agreement is possible with few players in a region to manage surplus scenarios
- For both FMCG and Cement company
 - Advertising and Branding can enable companies to realize a price premium if consumers are convinced of its superiority

- **Operating efficiency**

- For FMCG company
 - Distribution reach: to ascertain the geographical diversity of sales and ease in distributing new products
 - in-house production Vs. outsourcing
 - supply chain management: level of investment in IT enhancing inventory management and collection efficiencies
- For Cement company
 - Split-location plants: grinding units close to the consuming markets and the clinker unit near the limestone quarry
- For both FMCG and Cement company
 - efficiency in raw material sourcing;
 - flexibility in the manufacturing process
 - cost-cutting initiatives in use of manufacturing capacity
 - extent of integration, and research and development (R&D)
- **Quality of profitability vs. EPS growth**
 - For FMCG company growth depends on
 - Rising affordability of consumers
 - New products linked to consumer needs, values and lifestyles
 - For Cement company growth depends on
 - Level of government stimulus to rural / urban housing and infrastructure development
 - Private sector demand for infrastructure

For both FMCG and Cement company, rating agencies will focus on:

- **cash flow generation vs. book profitability** – policies on working capital management, credit agreements / contracts that limits extended credit period from suppliers and advances from customers/distributors
- **forward looking analysis** –projecting the possible future experience of the company
- **existing capital structure** (debt, equity, preference shares, short-term borrowings etc including holding company capital structure, where relevant)
- **financial flexibility** – unutilized bank / credit limits, liquid investments
- **accounting quality, debt servicing track record**

[1/2 mark for each bullet point]

Ratings will be formally evaluated at least once every 12 months and following any significant event like change in capital / business structure. [0.5]

Rating opinions are based on an evaluation of a company's financial strength, operating performance and market profile. Quantitative evaluation will typically involve assessment of performance over at least 5 years, using a battery of financial tests. [0.5]

Financial strength can be assessed in terms of:

- operating leverage = $\left(\frac{\text{sales} - \text{variable costs}}{\text{profit before interest \& tax}} \right)$
- financial leverage often referred to income leverage = $\left(\frac{\text{interest payments}}{\text{profit before interest \& tax}} \right)$

- asset leverage measured as $= \left(\frac{\text{debt}}{\text{debt} + \text{equity}} \right)$ or $\left(\frac{\text{debt}}{\text{equity}} \right)$.
- Asset quality, market value and diversification of assets, exposure to investment and credit risk.
- capital structure and liquidity (quick and current ratios, operational and net cash flows). [1]

The operating performance of a company can be assessed by considering its profitability (sources, trends) and revenue composition –sales of necessity/defensive goods, sales of luxury/cyclical goods, revenue from overseas subsidiaries, income from investments, exceptional items etc. [1]

Qualitative factors will include assessment of **management / promoter experience, strategy** (new projects / acquisitions / expansion), **risk appetite** and **objectives**, particularly with regard to the achievement of future business plans. The policies on leveraging; strength of other group companies; ability and willingness of group to capital infusion, if necessary [1]

[10]

ii) *Features of CBs, FDs and FMPs*

Credit Risk

- FMPs prospectus or offer for sale documents contain the investment objectives, intended asset allocation and the rating profile of the portfolio. This gives investor an idea of the returns and risks (e.g. credit risk).
- FDs can be considered as risk free.
- CBs come with the risk of default. Before investing in CBs checking the issuer's rating is a must. Credit rating is an indicator of the firm's ability to meet repayment obligations.

Reinvestment risk

- CB coupons need to be reinvested at unknown yields in future though this risk is not very significant over a four-year time horizon.
- To avoid this risk, the investor may opt for cumulative return option. There is no reinvestment risk in the FD and the FMP.

Marketability

- The marketability / liquidity of the CBs / FD / FMPs is often low but is not a concern as four-year time horizon matches with the maturity of all. FMPs usually trade at a discount to net asset value due to the management charges lowering the returns of underlying assets as held via FMP rather than directly.

Tax

- There may be considerable tax arbitrage that FMPs enjoy over CBs / FDs. Investors in FMPs get indexation benefits. CBs / FDs returns are taxed at the marginal income tax rate.

[1/2 mark for each bullet point]

[3]

iii) *Most suitable investment among CBs, FDs and FMPs*

Assumptions: Investment amount is one million currency units. To avoid reinvestment risk in CB, investor opts for cumulative return option. [0.5]

To allow for default risk in CB, gross yield reduced by 1%. This makes CB and FD identical. [1]

Value of CB / FD at maturity:

$$= 1,000,000 \left(1 + ((1 + 8\%)^4 - 1)(1 - 30\%) \right)$$

$$= 1,252,342.27 \quad [1.5]$$

Value of FMP at maturity

$$= 1,000,000 \left((1 + 7\%)^4 - ((1 + 7\%)^4 - (1 + 4\%)^4)(20\%) \right)$$

$$= 1,282,608.52 \quad [1.5]$$

Most suitable investment is FMP as it provides highest maturity value. [0.5]

[5]

iv) Recent news of FMPs payments withheld

Few FMPs of some mutual fund houses that were maturing have withheld payments (not paid the investors in full, or have extended the maturity date). Potential reasons are:

- they had exposure to companies that defaulted in redeeming loans / bonds [1]
- promoter funding: the collateral was the shares of the same companies [0.5]
- as many lenders exit would crash prices leading to minimal recovery [1]

In FMPs, indicative return does not mean assured return and upon significant defaults may give negative returns. As he is risk-averse, he should invest in FDs. [1]

[3]

[21 Marks]

Solution 4:

i) Behavioral Finance arguments

Anchoring and adjustment - While entering the trade he might have set a target (< -2 % and stop loss > 1 %) for example, say 3% fall in future value due to warning letter. This acts as anchor and in spite of evidence to the contrary, not adjusting his estimate adequately. [1]

Over confidence might be reinforced by **confirmation bias** - people will tend to look for evidence that confirms their point of view and tend to dismiss that does not justify it, in this case persistently discount any good news coming about ABC company. [1]

Regret aversion-The trader may fear that the future may be closed out just before the price begins to fall more. This would cause greater psychological pain than simply holding the future little longer. [1]

Status quo bias- Status quo bias refers to the fact that people have a marked preference for keeping things as they are. This is related to regret aversion. [0.5]

Trader might have been prone to **herding** behavior. His co-traders might have taken and holding on to the same trade and he desires to conform. [1]

[3]

ii) Steps to mitigate losses

Steps	Initial cost / capital required	At maturity of the short future	
		Share price goes up loss from short future	Share price goes down gain from short future
Buy one lot call option	Small cost of premium on call option	capped depending on premium paid and strike price of the call option	offset to the extent of premium paid
Sell one lot put option	Substantial capital for the margin requirements	offset to the extent of premium income on the put option	capped depending on premium income and strike price of the put option
Buy shares equal to one lot size	Full capital (or margin) to buy shares	offset to some extent by profits from sale of shares	shares can be held until making modest profits and enjoying any dividends in between
Long one lot future maturing in two / three months	Substantial capital for the margin requirements	offset to some extent from profits by closing out the long future position	long future position can be closed out or rolled over until making modest profits

For first two steps, options term should be matching that of the short future. For last two steps, if ABC share price goes down in this month then there is a risk of ABC shares never rising up to the desired level to sell shares / close long future.

[2 marks/ row]

[4]

[7 Marks]

Solution 5:

i)

a) FTSE 100 Index:

This consists of the 100 largest quoted companies by market capitalisation, accounting for about 80 per cent of the total UK equity market capitalisation.

It is the main indicator of short-term movements in the UK equity market and is used extensively as a basis for investment products, such as derivatives and exchange traded funds. The capital value is the one that is widely quoted.

The index is calculated on a real-time basis, ie continuously during the day, effectively every minute, and is based on the last trade prices.

For continuity and administrative reasons, the constituents are changed once a quarter in accordance with predetermined rules.

The index is also used as a basis for stock index derivatives, namely the FTSE 100 Index Futures and the FTSE 100 Index Options. [3]

b) Chicago Board Option Exchange Volatility Index:

This index measures the volatility of the US equities.

This is typically used as an indication of the market perception of risk.

This is essentially a weighted average of prices for a range of 30 day expiry put and call options on the S&P 500 index.

Intuitively, the level of the index, which is quoted in annualized percentage terms, is the risk neutral expectation of the volatility on the S&P index over a 30 day period.

The composition of VIX uses the prices of options to calculate the implied volatility.

Specifically, a higher option price implies greater volatility, other things being equal. [3]

c) Markit iTraxx Europe Index:

This is the index of credit default swaps.

This can be used by analysts and investors to monitor more directly the price of credit risk.

Different index series are established each six months, and the constituents of the index are the top 125 names in terms of CDS volume traded in the six months prior to this.

The index is quoted for 3, 4, 5, 7 and 10-year maturity CDS and the index is the unweighted average of the CDS premium for the constituents.

The constituents of the indices are changed every six months, a process known as "rolling" the index.

There are also several sub-sector indices for different industries. [3]

ii)

Global index series covers majority of the world's investible market capitalization whereas local indices would be confined to that market. [1]

There might be some stocks which are not available to foreign investors. The local indices might include the stocks. Whereas, the Global index series exclude such stocks and would be more suitable. [1.5]

The local indices might not be sufficiently comprehensive or calculated consistently and accurately. [1]

Global index series have the advantages of being consistent between countries and are easier to obtain than some local indices. [1]

In Global index series, index numbers are shown for each country in US dollar and local currency terms. The local currency index gives a measure of the underlying performance of the particular market, and the dollar currency index shows performance adjusted for movements in the currency concerned. [2]

The local indices might not give any indication for movements in the concerned currency. [1]

[6]

[15 Marks]

Solution 6:

- i) The policy account represents the balance amount to the policyholder. The assets are invested majorly in Government securities and Corporate Bonds. [1]
 In case of rise in bond yields, the market value of assets would decrease while the liability wouldn't. [1]
 In such case if a policyholder wants to surrender or withdraw significant amount of its fund, then the Company would have to liquidate the assets at a lower value while it has to pay the full policy account value and thus it would incur loss. [2]
[4]

- ii) The Company can add market value adjustment (MVA) in its policy conditions in case there is significant withdrawal (bulk exit) or complete surrender. [1]

The Company would have to define bulk exit. For example, it can term any exit of more than 25% of the policy account as bulk exit. [1]

There might be some policies where the policy account is in thousand of crores. In such cases, a threshold of 25% would still expose the Company to significant losses. [1]

So, the Company can also define bulk exit in absolute amount say any exit more than 100 crore in addition to the above definition. [0.5]

The Company would have to define the formula by which MVA can be applied. [0.5]

A formula could be to apply the ratio of current market value to the current account value subject to a maximum of 1. [1]

The Company can also describe the economic situations in which it would apply MVA. [0.5]

For example – if the current G-sec rate is 100 basis points higher than the last one-year average. [0.5]

By applying MVA, the Company can pay lesser than the current policy account value. [0.5]

This would safeguard the Company against losses in case of increase in bond yields. [1]

The MVA might not be appreciable by the clients. So, the Company can also offer to pay the withdrawal amount in instalments over some period so that it has time to liquidate the assets and can limit its losses. [2]
[8]

- iii) The modelling can be done deterministically or stochastically. [0.5]

In a deterministic framework, we can decide the nature and extent of the scenarios to be tested for the purpose of setting the reserves. [1]

As majority of the assets are invested in Government securities and Corporate Bonds, so we might consider the impact of immediate increase of 100 basis points in bond yields. [2]

The value of assets and liabilities would be compared. [0.5]

Asset liability mismatch reserves can then be set up to cover the possible levels of shortfall identified. [1]

In a stochastic framework, we can make multiple projections to generate many possible future economic scenarios. [0.5]

The stochastic element of the projections would apply to the asset portfolio and investment returns, in order to assess exposure to systematic risk. [1]

The assessment of the results could be carried out in the form of ruin probability. That is, the outcomes are ranked in terms of a target measure (such as the shortfall of assets relative to liabilities at a specified future date). [2]

Additional reserves are then set up at a level sufficient to cover all but a specified proportion of such shortfalls. [1]

For example, suppose that the 10th worst outcome out of 1,000 simulations produced a shortfall of \$10 million. This suggests that an extra \$10 million of reserves would be required if the aim of the investor is to ensure that the probability of ruin is no greater than 1%. [1.5]
[8]

[20 Marks]

Solution 7:

- i) **Market Risk:** the risk relating to changes in the value of the portfolio due to movements in the market value of the assets held.

- Credit risk** is the risk that a counterparty to an agreement will be unable or unwilling to fulfil their obligations.

- Operational risk** is the risk of loss due to fraud or mismanagement within the fund management organisation itself.

- Liquidity risk:** The risk of not having sufficient cash to meet operational needs at all times. It is related to market risk in as much as the liquidity of the overall portfolio is need to be taken into account in portfolio selection.

- Relative performance risk** is the risk of under-performing comparable institutional investors. [5]

- ii) First, the set of asset classes that could be included in the portfolio needs to be defined. Estimates of the volatilities and covariances of each asset class needs to be considered. Use asset-liability modelling to determine the matching portfolio. VaR calculations would need to be performed to determine the total risk budget. The total risk budget needs to be allocated between strategic risk and total active risk.

Further modelling of investment risks and returns to investigate the different possible investment strategies.

Allocating the total active risk between the different investment managers

Choosing the investment managers and allocating their benchmarks.

Monitor risk exposures (increases and decreases in the values of the positions) and changes in volatilities and correlations.

Rebalance the portfolio in response to changes in the short-term conditional volatility and correlations of the assets.

Allocations are altered to keep the overall portfolio risk at the level defined as tolerable for the investor.

[8]

- iii)** This can free the manager to look for alternative investments that might increase the expected return on the portfolio.

Because the constraint is that the total risk of the portfolio must stay at or below a targeted level, increased attention is paid to low correlation investments.

Small allocations to such investments can reduce the total risk of the portfolio through diversification.

[2]

[15 Marks]
